Resolution Geometry: The Physical Shortcut Lock

What Ordinary Matter Can Shorten, What It Cannot, and How to Certify It

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Abstract

We define a shortcut operationally. A route has an *anchored advantage* over a reference when its certified upper arrival bound lies below the reference's certified lower bound. Both are taken for the same departure event, the same destination worldline, the same clock and the same launch energy. Under this definition, with the Einstein coupling declared and sources that satisfy the null energy condition (NEC), we lock three results.

- **Light cannot be advanced.** Kinetic sources only delay null rays (no-go).
- **Slow travellers can be advanced.** At first order the transit rate is

$$\boxed{\frac{dt}{dl} = \frac{1}{v}\left[1 + |\Phi|\left((2+\gamma) - \frac{1}{v^2}\right)\right]}$$

  so a positive-mass well makes travellers with $v^2 < 1/(2+\gamma)$ arrive *earlier* than on the flat route with the same launch energy. Faster travellers are delayed. Exact integration of the declared metric confirms the sign law at 16 speeds. The flip speed $v_c = 1/\sqrt{2+\gamma}$ is recovered to 10^{-5}, so a two-speed transit panel measures γ.
- **Gravity trains need no propulsion.** A chord tunnel through a uniform sphere delivers any surface point to any other in $\pi\sqrt{R^3/GM} = 42.17$ min for Earth, independent of the chord.

The uploaded coverage-control package certifies crossing and arrival windows through a declared throat, with bounds that always contain the true arrival. Its source fails the NEC, and its global comparison is unresolved.

So a physical shortcut *exists and locks* in its ordinary-matter form: geometry converts gravitational potential into arrival advantage for slow travellers and propulsion-free transit. A shortcut that beats light remains refused under the NEC. The gates it would have to break are listed with their current status.

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1 The anchored definition

Definition 1.1 (Anchored advantage)**.** Fix four things: a departure event p, a destination worldline Γ with its clock, a launch-energy budget, and a reference route. A route has a certified *anchored advantage* if

$$T^{\text{upper}}_{\text{route}} < T^{\text{lower}}_{\text{reference}},$$

where both bounds are arrival readings on Γ's clock and include every declared modelling and coverage error.

This definition removes three failure modes of earlier attempts:

- comparing different clocks;
- comparing a throat transit with an unspecified exterior route;
- comparing a short optical time with a traveller who turns back.

2 Light: no advance

Imported Theorem 2.1 (No null advance from kinetic sources)**.** *In linearized Einstein gravity, the coordinate-time delay of a null ray k is*

$$\Delta t = 2G \int dl \int \frac{T_{ab}k^a k^b}{|x - x'|}\, d^3x'.$$

RG's kinetic sources satisfy $T_{ab}k^a k^b = \int (p \cdot k)^2\, d\mu \geq 0$. Hence $\Delta t \geq 0$: kinetic matter never advances light. The battery `rg_no_shortcut.py` *checks 200 random sources. In full GR the same conclusion holds under the NEC (Gao–Wald [1]), and traversable wormholes are excluded by topological censorship [2].*

3 Massive travellers: the transit sign law

Theorem 3.1 (Transit sign law)**.** *Take the metric $ds^2 = -(1 + 2\Phi)\, dt^2 + (1 - 2\gamma\Phi)\, dl^2$ with $\Phi \leq 0$, and a straight geodesic route. Fix the conserved energy $E = 1/\sqrt{1 - v^2}$, i.e. the launch speed v in the flat region. To first order in Φ,*

$$\frac{dt}{dl} = \frac{1}{v}\Big[1 + |\Phi|\Big((2 + \gamma) - \frac{1}{v^2}\Big)\Big].$$

The route through the well therefore arrives earlier than the flat route with the same launch energy iff

$$v^2 < \frac{1}{2 + \gamma}.$$

Proof. Energy conservation gives $\dot{t} = E/(1 + 2\Phi)$. Normalization gives $(dl/dt)^2 = \frac{1+2\Phi}{1-2\gamma\Phi}\big[1 - (1 + 2\Phi)(1 - v^2)\big]$. Expanding to first order,

$$(dl/dt)^2 = v^2 - 2\Phi + (4 + 2\gamma)\Phi v^2,$$

and inverting gives the stated rate. □

Meaning. The two terms are the two channels of the duality paper:

- the energy channel (the lapse N) speeds slow travellers up as potential energy becomes kinetic energy;
- the structure channel (h, weighted by γ) lengthens every path.

For light the structure term wins, giving the Shapiro delay $(1+\gamma)$. For slow travellers the energy term wins, giving an advance. Nothing exotic is required: the advance is paid for by the gravitational potential and returned on exit, and NEC-respecting matter is enough.

v	excess, exact metric	excess, first order	sign
0.05	-9.371	-9.514	advance
0.2	-0.1317	-0.1318	advance
0.4	-0.00973	-0.00974	advance
0.557	-4.7×10^{-4}	-4.7×10^{-4}	advance
0.597	$+4.0\times10^{-4}$	$+4.0\times10^{-4}$	delay
0.95	$+0.00239$	$+0.00239$	delay

The table uses $\gamma=1$, $GM=10^{-4}$, a Plummer-softened well of scale 1, and a route of length 400 ($c=G=1$). The second-order geodesic ODE agrees with the quadrature to 10^{-6}. The case $\gamma=0$ also checks: its sign flips at $1/\sqrt{2}$.

Corollary 3.2 (A two-speed panel measures γ; P3)**.** *The flip speed $v_c=1/\sqrt{2+\gamma}$ is recovered by exact integration as* 0.707098*,* 0.632446 *and* 0.577340 *for $\gamma=0$,* 0.5 *and* 1*; the predictions are* 0.707107*,* 0.632456 *and* 0.577350*. Timing transits at two speeds on either side of v_c therefore measures the structure channel directly, as the duality theorem requires.*

4 Propulsion-free transit: the gravity train

Proposition 4.1 (Chord transit; P4)**.** *Assume a uniform-density sphere and an evacuated, frictionless chord tunnel. Released from rest, a carriage performs simple harmonic motion along the chord and arrives at the far end after $\pi\sqrt{R^3/GM}$, for every chord. For Earth's parameters this is* 42.17 *min.*

Matching that arrival along the surface needs a sustained speed of 0.44, 2.64, 5.27 and 7.91 km/s for separations of 10°, 60°, 120° and 180°. The antipodal value is orbital speed. GR corrections are of order $GM/(Rc^2)\sim 7\times10^{-10}$. A non-uniform Earth shortens the antipodal time [3]. Tunnelling, vacuum, friction and heat are engineering gates and are not addressed here.

5 Certificates under coverage

Proposition 5.1 (Anchored certificate; P5)**.** *At $v=0.2$, the route through the well has upper bound* 1999.901 *even after a coverage budget of* 25% *of the effect is charged to both sides. The flat reference with the same launch energy has lower bound* 1999.967*. The advantage is* ***certified****. At $v=0.95$ the route is slower, and the certificate is* ***refused****.*

Imported Theorem 5.2 (Coverage control (uploaded package))**.** *Suppose lapse samples carry error η and a derivative bound K_1 holds. With spacing h,*

$$\Phi_{\max}\le \max y_i+\eta+K_1h/2,$$

and $E_{\min}>e^{\Phi_{\max}}$ certifies crossing. For the declared throat ($a=1$, $\beta=-1$, $v=0.95$) the destination-clock bounds tighten as follows, and every bound contains the integrated 9.0187*:*

- *17 cells:* $[7.67, 11.09]$*;*
- *32 cells:* $[8.22, 10.07]$*;*
- *128 cells:* $[8.79, 9.27]$*;*
- *512 cells:* $[8.945, 9.093]$.

Tidal safety needs an additional bound K_2 on Φ''; sampling alone cannot supply it. A hidden compact barrier between samples defeats any certificate without K_1.

6 The throat: conditional traversal only

The Ellis–Bronnikov throat is crossable, and its arrival can be bounded under coverage. However, the declared Einstein equations require $\rho + p_r = -2/a^2$ at the throat, which kinetic matter cannot supply. Its global anchored comparison with an exterior route is undefined without a same-exterior mouth placement.

The negative null energy a throat of radius a needs is about $c^2/(4\pi G a^2)$:

Throat radius	Required deficit (kg/m^3)
1 m	1.1×10^{26}
1 km	1.1×10^{20}
1 AU	4.8×10^{3}
1 light-year	1.2×10^{-6}
Hubble radius	5.7×10^{-27}

For comparison, the phantom margin hinted at by recent dark-energy fits is about 10^{-27}–10^{-28} kg/m^3 [4].

7 Gate ledger for a shortcut that beats light

Gate	Status	RG probe
G1 Coupling	locked to about 10^{-5} (Cassini γ)	two-speed transit panel (this paper); duality
G2 Laboratory null energy	kinetic matter: no advance (theorem); quantum: local negative energy, averaged positivity in flat space	time-advance certificate: an observed advance certifies negative null energy on the line of sight
G3 Cosmic null energy	open: DESI DR2 combinations favour a phantom crossing [4]	saturation budget (Section 6)
G4 Route coverage	local probes insufficient (hidden barrier)	full-coverage certificates, or abstain
G5 Identification and control	conditional lock for the declared throat	two channels, capped policy

8 Verdict

- **Locked, physical, ordinary matter:**
  - positive-mass wells advance slow travellers relative to the same-energy flat route;

  - gravity trains deliver propulsion-free transit;
  - both carry anchored certificates.

  These are real shortcuts. Geometry converts potential into arrival advantage with no propellant and no exotic matter.

- **Refused under the NEC:** any advance over *light*. Opening it requires breaking G2 or G3. RG supplies the instruments that would detect such a break (the time-advance certificate) and measure its distance from feasibility (the saturation budget).
- **Conditional:** throat traversal under coverage, with the source unresolved.

9 Battery, limits and non-claims

- `rg_physical_shortcut.py` (NumPy, SciPy): P2 sign law at 16 (speed, γ) cases; ODE cross-check; P3 flip speeds; P4 chord independence; P5 certificate and refusal.
- `rg_no_shortcut.py`: P1.
- The uploaded coverage-control results are audited: every refinement bound contains the integrated arrival.

Limits. The calculations are first order and weak field, with test particles and a declared metric. The gravity train assumes uniform density and no friction. No experiment has been performed. The P5 coverage budget is declared, not measured.

Non-claims. No faster-than-light travel. No violation of physical law. The advance for slow travellers is ordinary gravitational acceleration, typed as an anchored arrival advantage. It does not beat light.

References

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